

DEBARSHI ROY

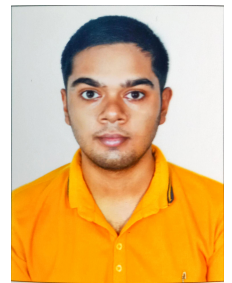
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Kolkata, West Bengal - 700030, India

DOB: 30/06/2000 | Gender: Male | Citizenship: Indian

International PhD Applicant (Fall 2027 Intake)



RESEARCH INTERESTS & MOTIVATION

Research Interests: Algorithms and Computer Architecture for Biomedical Computing · AI/ML for Computational Metagenomics/Genomics · Domain-Specific Accelerators for Computational Metagenomics/Genomics Tools · High-Performance Computing for Computational Biology · Hardware-Software Co-design

Motivation: Inspired by early mentorship in high school biology, my interest in scientific research has evolved into a focused research vision at the intersection of computer architecture, algorithms, and computational biology. As a prospective PhD candidate, I aim to develop high-performance, AI-driven bioinformatics systems that address critical computational bottlenecks in clinical metagenomics. My goal is to bridge advances in algorithms and hardware to enable biological discovery, transforming massive, complex genomic/metagenomic datasets into scalable, efficient, and clinically actionable diagnostic solutions.

EDUCATION

Indraprastha Institute of Information Technology Delhi

Master of Technology in Computational Biology

• CGPA: 7.38/10.00

August, 2025 - July, 2027

New Delhi, Delhi, India

Maulana Abul Kalam Azad University of Technology

Bachelor of Technology in Computer Science & Engineering

• CGPA: 8.80/10.00

June, 2019 - June, 2023

Kolkata, West Bengal, India

PATENTS AND PUBLICATIONS

C=Conference, J=Journal, B=Book Chapter, P=Patent, S=In Submission, T=Thesis

- [B.1] **Debarshi Roy**, Soumita Banerjee, Alisha Ansari, Sourav Goswami, Nidhi P. Raval, Kaustav Adhikary, and Tarini Shankar Ghosh (2026). **Artificial Intelligence (AI) in microbiome-directed biotherapeutics development**. *Progress in Molecular Biology and Translational Science*, Elsevier Inc. DOI: [10.1016/bs.pmbts.2026.01.005](https://doi.org/10.1016/bs.pmbts.2026.01.005)
- [J.1] **Debarshi Roy**, Nidhi P. Raval, Vindhya Regonda, Omprakash Shete, Kaustav Adhikary, Shivangi Verma, and Tarini Shankar Ghosh (2026). **Overlapping signatures of gut microbiome in aging and disease: Enrichment of medication-associated oral-microbiome-predominant microbes in the gut**. *FEBS Letters*. DOI: [10.1002/1873-3468.70391](https://doi.org/10.1002/1873-3468.70391)

PROJECTS

METAQR: Image-Based Analysis in Microbiome

at IIIT Delhi

Tools: [R/Python, Flet, HTML/JS, Git]

- Engineered a scalable clinical diagnostic tool that ultra-compresses massive metagenomic profiles (15 GB) into lightweight, byte-sized patient signatures.
- Developed and deployed a serverless, cross-platform application (Web, iOS, Android) utilising dynamic QR codes for instantaneous point-of-care visualisation of gut ecology.
- Processed over 70,000 patient samples to construct a highly optimised, low-memory clinical database, resulting in a formal patent filing for the system architecture.

NEMESIS: Telemetry-Driven Distributed LLM Scheduler 

at IIIT Delhi

Tools: [Python, Kubernetes, PyTorch DDP, Docker, Linux tc, Bash]

- Engineered a custom, multi-objective Kubernetes scheduler that intercepts default container placement to dynamically route LLM workloads, reducing LLM execution time by 62% under controlled CPU-contention workloads.
- Implemented a topology-aware gang-scheduling algorithm that uses inter-rack latency measurements to preserve efficient PyTorch DDP communication across distributed GPU workloads.
- Integrated real-time NVIDIA NVML telemetry to bypass bare-metal GPU VRAM fragmentation, cutting compute times by over 40% and isolating network bottlenecks via deterministic latency sweeps.

ELECTRA-Based Microbiome Representation Learning

at IIIT Delhi

Tools: [Python, PyTorch, Transformers, Pandas, SciPy]

- Designed a predictive framework leveraging the ELECTRA language model architecture to interpret and extract unstructured patterns from human microbiome metadata.
- Adapted natural language processing pre-training objectives (replaced token detection) to model complex biological sequences, generating robust embeddings for disease-specific prediction tasks.
- Built scalable data processing pipelines to tokenize high-dimensional microbial profiles, aiming to improve computational efficiency and baseline diagnostic accuracy.

RESEARCH EXPERIENCE

Indraprastha Institute of Information Technology Delhi

May, 2025 - July, 2025

Research Intern in the Microbiome Informatics Lab of the Computational Biology Department

New Delhi, Delhi, India

- Developed a predictive framework leveraging large language models (LLMs) to model disease-specific patterns in human microbiome metadata.

Indian Institute of Technology Guwahati

January, 2024 - December, 2024

Former Research Assistant in the MARS Lab of Computer Science & Engineering Department

Guwahati, Assam, India

- Teaching Assistant of the course "**Introduction to Computing**" (CS 101), Course Instructors: Prof. T. Venkatesh & Dr. John Jose, from January, 2024 - May, 2024.
- Teaching Assistant of the course "**Digital Design & Computer Architecture**" (CS 322M), Course Instructor: Prof. Jatindra Kumar Deka, from July, 2024 - November, 2024.
- **Experience Certificate**

Indian Institute of Science Education and Research Bhopal

July, 2023 - December, 2023

Former JRF in the Electrical Engineering and Computer Science Department

Bhopal, Madhya Pradesh, India

- Teaching Assistant Position of the course **Discrete Mathematics (ECS 201)**, Course Instructor: Dr. Kundan Kandhway.

TECHNICAL SKILLS

Programming Languages: Python, R, C, C++

Systems & Architecture: Linux, CUDA, PyTorch DDP, Kubernetes, Docker, gem5

Scientific Computing: NumPy, SciPy, Pandas, WGCNA

ADDITIONAL INFORMATION

Languages:

- English (Professional working proficiency)
- German (Professional working proficiency)
- Hindi (Limited working proficiency)
- Bengali (Native or bilingual proficiency)

Hobbies: Cricket, Chess, Swimming, Drama

REFERENCES

Dr. Tarini Shankar Ghosh

Assistant Professor, Department of Computational Biology

Indraprastha Institute of Information Technology Delhi, New Delhi-110020, India

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Relationship: MTech Thesis Supervisor in IIIT Delhi

Dr. Rinku Shah

Assistant Professor, Department of Computer Science & Engineering

Indraprastha Institute of Information Technology Delhi, New Delhi-110020, India

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Relationship: Project Guide & Course Instructor in Graduate System Course in IIIT Delhi

Dr. N. Arul Murugan

Associate Professor, Department of Computational Biology

Indraprastha Institute of Information Technology Delhi, New Delhi-110020, India

Email: arul.murugan@iiitd.ac.in

Relationship: Project Guide & Course Instructor in Algorithms for Computational Biology Course in IIIT Delhi